

Lion Guardians Collared Report

Technical Guide

Collared Lion Movement Analysis — Methodology & Calculation Reference
Version 1.0 · Generated June 08, 2026

1. Overview

The **LG Collared Report** workflow analyses GPS-collared lion movement in the Amboseli ecosystem for the **Lion Guardians** programme. It ingests telemetry from **EarthRanger**, computes home range and track maps per subject, generates speed and distance statistics, and delivers an interactive dashboard plus a print-ready Word report.

Note: All per-group outputs (maps, metrics, Word sections) are produced by iterating over a user-chosen grouper: subject name, sex, or subtype.

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All telemetry is fetched from an **EarthRanger** instance via `set_er_connection`. The workflow calls `get_subjectgroup_observations` once over the configured time range. The filter: `clean` parameter discards junk-flagged observations before they reach the workflow.

2.2 Groupers

Three grouper fields are available via `set_groupers`:

Grouper field	EarthRanger source	Typical use
subject_name	subject.name	One output per individual collared lion (default)
subject_sex	subject.sex	Aggregate by sex (M / F)
subject_subtype	subject.subject_subtype	Aggregate by subtype

2.3 Static Geodata Files

Two boundary datasets are downloaded from Dropbox and cached locally:

Dataset	File	Purpose
Group Ranch Boundaries	lg_group_ranch_boundaries.gpkg	Community ranch polygons in Amboseli
Conflict Hotspot Areas	lg_conflict_hotspots.gpkg	Known human–lion conflict hotspot features

Both files use `overwrite_existing: false` (3 retries). After loading, each is reprojected to **EPSG:4326** and annotated with its geometry type before layer creation.

2.4 Word Document Templates

Template file	Purpose
collared_lions_cover_page.docx	Report cover page — subject count, time range, preparer
collared_lion_subject_template.docx	Per-grouper section — metrics and map images

2.5 Base Map Tile Layers

Layer	Opacity	Max zoom
ArcGIS World Hillshade	100 %	20
ArcGIS World Street Map	10 %	20

The hillshade provides full-opacity terrain context. The street map is overlaid at 10 % to show road networks and settlement names without obscuring the hillshade or data layers.

3. Data Ingestion Pipeline

3.1 Observations → Relocations

`process_relocations` converts raw EarthRanger observations to a standardised GeoDataFrame. Retained columns:

Column	Source field	Description
<code>groupby_col</code>	<code>internal</code>	Subject identifier for grouping
<code>fixtime</code>	<code>observation timestamp</code>	UTC datetime of the GPS fix
<code>junk_status</code>	EarthRanger flag	True if fix is marked junk
<code>geometry</code>	<code>lat/lon</code>	Point geometry (WGS 84)
<code>extra__subject__name</code>	<code>subject.name</code>	Lion display name
<code>extra__subject__subject_subtype</code>	<code>subject.subject_subtype</code>	Subject subtype
<code>extra__subject__sex</code>	<code>subject.sex</code>	Subject sex (M / F / unknown)

Three bad coordinate pairs are removed unconditionally:

- (180.0, 90.0) — boundary sentinel
- (0.0, 0.0) — null-island artefact
- (1.0, 1.0) — common default / test value

Cleaned relocations are persisted as `relocations.geoparquet`.

3.2 Relocations → Trajectories

`relocations_to_trajectory` connects consecutive fixes per subject into `LineString` segments, adding `dist_meters`, `speed_kmph`, `segment_start`, and `segment_end`. A `trajectory_segment_filter` removes biologically implausible segments before analysis:

Filter parameter	Default	Description
<code>min_length_meters</code>	10	Discard segments shorter than 10 m
<code>max_length_meters</code>	10 000	Discard segments longer than 10 km
<code>min_time_secs</code>	10	Discard segments shorter than 10 s
<code>max_time_secs</code>	21 600	Discard segments longer than 6 hours
<code>min_speed_kmph</code>	1	Discard segments below 1 km/h average speed

Filter parameter	Default	Description
max_speed_kmh	30	Discard segments above 30 km/h average speed

Trajectories are persisted as trajectories.geoparquet.

3.3 Temporal Index & Speed Classification

`add_temporal_index` keys the trajectory GeoDataFrame to `segment_start`, grouped by the configured groupers, enabling per-group iteration. `apply_classification` then bins `speed_kmh` into **6 equal-interval classes** (output column: `speed_bins`, labels to 1 decimal place).

3.4 Column Renaming & Group Splitting

Three columns are renamed via `map_columns` (`raise_if_not_found: true`):

Original column	Renamed to
extra__name	subject_name
extra__sex	subject_sex
extra__subject_subtype	subject_subtype

The renamed GeoDataFrame is split into per-group partitions by `split_groups`. All downstream map and metric tasks iterate over these partitions via `mapvalues`.

4. Static Map Layers

Three static layers are built once and composited onto every subject-level map to provide spatial context.

4.1 Layer Styles

Layer	Colour (RGB)	Opacity	Filled	Notes
Group Ranch Boundaries	(169, 169, 169) grey	45 %	No	Outline only, line width 1.25
Conflict Hotspots	(220, 20, 60) crimson	45 %	Yes	Point radius 2.05, line width 1.25
Hotspot Text Labels	(20, 20, 20) near-black	—	—	Arial, 1 000 m base, 40–75 px clamp, centroid-anchored

5. Map Outputs — Methodology

5.1 Home Range Map (Elliptical Time Density)

`calculate_elliptical_time_density` computes an ETD kernel density estimate per group, accounting for temporal autocorrelation in GPS telemetry. Parameters:

Parameter	Value	Meaning
CRS	ESRI:53042	World Azimuthal Equidistant — equal-area, minimises distortion
percentiles	50, 60, 70, 80, 90, 95, 99	Contour probability thresholds extracted as polygons

Parameter	Value	Meaning
max_speed_factor	1.05	Estimated max speed as a multiple of the measured maximum; bounds the kernel spread
expansion_factor	1.3	Shape buffer expansion — controls how far density values spread across the grid
band_count	1	Single density band output
nodata_value	NaN	Cells outside probability mass are transparent

Contour polygons are coloured with the **RdYlGn** diverging colormap (innermost 50th percentile = red, outermost 99th = green). Map opacity is 45 %, legend title: *Home Range Percentiles*.

The map view is auto-zoomed via `envelope_gdf + custom_view_state_from_gdf` (pitch 0, bearing 0, max zoom 10). An interactive HTML is persisted (suffix: `homerange`) then converted to PNG via `adjust_map_zoom_and_screenshot` (2× device scale factor, 40 s tile-load wait).

5.2 Subject Tracks Map

`create_path_layer` renders all trajectory segments per group as a continuous blue path layer:

Property	Value
Colour	RGB(30, 144, 255) — dodger blue
Width	1.25 px, min 2 px, max 8 px (screen-space)
Cap / Join	Rounded
Opacity	45 %

The path layer is combined with the four static layers via `combine_deckgl_map_layers`. An interactive HTML is persisted (suffix: `tracks`) and converted to PNG using the same screenshot settings as the Home Range map.

6. Summary Metrics

6.1 Per-Group Summary Table

`summarize_df` aggregates trajectory data grouped by `subject_name`:

Output column	Source	Aggregator	Unit	D.p.
mean_speed	speed_kmhr	mean	km/h	2
min_speed	speed_kmhr	min	km/h	2
max_speed	speed_kmhr	max	km/h	2
total_distance	dist_meters	sum	m → km	2

`add_totals_row` appends a *Total* row across all subjects. The table is persisted as CSV.

6.2 Scalar Widget Values

Four values are derived per group via `dataframe_column_sum` then rounded to 2 d.p. by `round_off_values`:

- Mean Speed (km/h)
- Min Speed (km/h)
- Max Speed (km/h)
- Distance covered (km)

6.3 Unique Subject Count

`dataframe_column_nunique` counts distinct values in `groupby_col` on the temporally-indexed trajectory `GeoDataFrame`. This total is used in the cover page context.

7. Word Report (.docx)

7.1 Cover Page

`create_cl_ctx_cover` builds the cover context (subject count, time range, *Ecoscope* as preparer). `create_context_page_lg` renders it into `lg_cover_page.docx`.

7.2 Per-Grouper Sections

`create_collared_lions_grouper_ctx` assembles a context dict per group containing the trajectory `DataFrame`, total distance, Home Range PNG, and Tracks PNG. `create_grouper_page` renders each section from the subject template. Image boxes: **11.11 × 6.5 cm**. `validate_images: true` catches missing PNGs before rendering.

7.3 Document Merge

`merge_cl_files` concatenates the cover page and all per-grouper sections into a single Word file saved to the results directory.

8. Interactive Dashboard

`gather_dashboard` assembles the **Lion Guardians** dashboard from six widget groups:

Widget	Type	Source
Mean Speed	Single value	<code>round_mean_speed</code> → <code>total_mean_speed_sv_widget</code>
Min Speed	Single value	<code>round_min_speed</code> → <code>total_min_speed_sv_widget</code>
Max Speed	Single value	<code>round_max_speed</code> → <code>total_max_speed_sv_widget</code>
Distance covered	Single value	<code>round_total_distance</code> → <code>total_distance_sv_widget</code>
Home Range Metrics	Map	<code>td_ecomap_html_url</code> → <code>td_grouped_map_widget</code>
Subject tracks	Map	<code>track_html_url</code> → <code>sm_grouped_map_widget</code>

Note: Widget tasks use `skipif: [never]` so the dashboard always assembles, even when some groups have no data.

9. Output Files

All files are written to `$ECOSCOPE_WORKFLOWS_RESULTS`.

File / pattern	Format	Content
relocations.geoparquet	GeoParquet	Cleaned GPS fix locations
trajectories.geoparquet	GeoParquet	Segments with speed_kmhr, dist_meters, speed_bins
_homerange.html	HTML	Interactive ETD home range map
_tracks.html	HTML	Interactive subject tracks map
_homerange.png	PNG	2× screenshot of home range map
_tracks.png	PNG	2× screenshot of tracks map
_summary.csv	CSV	Speed and distance summary table with totals row
lg_cover_page.docx	Word	Rendered report cover page
.docx	Word	Per-grouper report section
.docx	Word	Final combined report

10. Workflow Execution Logic

10.1 Skip Conditions

Two default skip conditions apply to every task (task-instance-defaults):

- **any_is_empty_df** — skips the task (and all dependants) when any input DataFrame is empty, handling subjects with no fixes gracefully.
- **any_dependency_skipped** — propagates skips downstream automatically.

Widget and map-widget tasks override this with skipif: [never] to ensure the dashboard always assembles.

10.2 Data Flow Summary

Stage	Tasks
Setup	EarthRanger connection, time range, groupers, base maps
Geodata download	2 boundary files + 2 Word templates from Dropbox
Static layers	Ranch, hotspot, hotspot text layers
Telemetry ingest	Observations → relocations → trajectories → speed bins → rename → split
Home Range branch	ETD → colormap → map layer → compose → view state → HTML → PNG → widget
Tracks branch	Path layer → compose → draw map → HTML → PNG → widget
Metrics branch	Summary table → totals row → CSV → scalar widget series
Report assembly	Cover page + per-group sections → merge docx
Dashboard	gather_dashboard combines all widgets

11. Software Versions

Package	Version	Role
ecoscope-workflows-core	0.22.17.*	Core task library and workflow engine
ecoscope-workflows-ext-ecoscope	0.22.17.*	Spatial analysis tasks (ETD, relocations, trajectories)
ecoscope-workflows-ext-custom	0.0.40.*	Utility tasks (column mapping, rounding, screenshots)
ecoscope-workflows-ext-ste	0.0.18.*	Summary table and totals-row tasks
ecoscope-workflows-ext-mnc	0.0.7.*	MNC domain tasks
ecoscope-workflows-ext-big-life	0.0.8.*	Big Life Foundation domain tasks
ecoscope-workflows-ext-lion-guardians	0.0.6.*	Lion Guardians Word report rendering tasks

Packages are distributed via the `prefix.dev` conda channel and pinned to patch-compatible versions (*. suffix). The runtime environment is managed by **pixi**.